COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 03

Miscellaneous Topics

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Recap of Previous Lecture



- Number System.
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Number System Cont.

Q. A quadratic equation in base-r system is given as:

$x^2 - 18x + 80 = 0$

If one of the root of this equation is $x = (8)_r = (8)_{10}$ then value of r is:

a. $r = 10$

b. $r > 8$

c. $r > 7$

d. None of these.

$r \geq 9$

$(18)_r = (r+8)_{10}$

$(80)_r = (8r)_{10}$

in decimal no system:

$x^2 - (r+8)x + 8r = 0$

$x = 8$ is a solution then

$64 - (r+8) \times 8 + 8r = 0$

↳ satisfies for any real value of r

$x(x-8) - r(x-8) = 0$

$(x-8)(x-r) = 0$

$x = (8)_{10}, (r)_{10}$ ✓

$r \geq 9$ ∩ $r > 8$ overall

$r \geq 9$ ≡ $r > 8$

$n = 9$

$$(x^2 - 17x + 72)_{10} = 0$$

$$(x-9)(x-8) = 0 \Rightarrow \boxed{x = 9, 8} \Rightarrow$$

$$(10)_9$$

$n = 11 \longrightarrow \quad x^2 - 19x + 88 = 0$

$$(x-11)(x-8) = 0$$

$$x = (8)_{10}, (11)_{10}$$

$$(10)_{11}$$

b. $(x^2 - 15x + 50)_r = 0$

if one of the root $x = (10)_r$

then minimum value of r is ——.

H.W.

Question

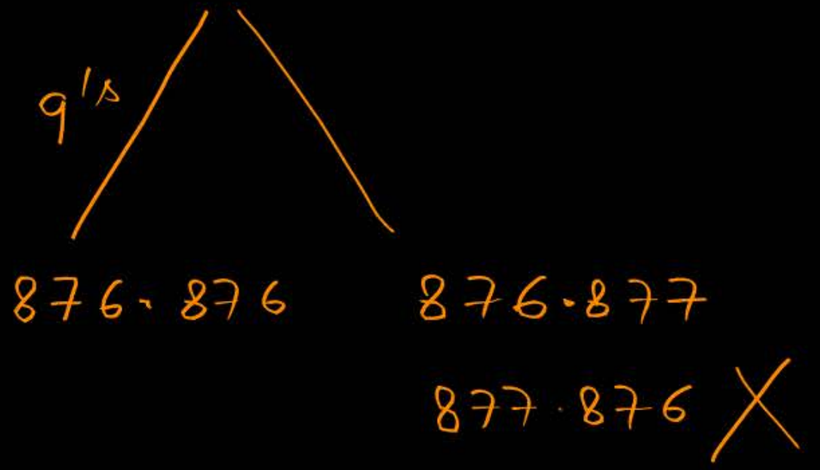
9's and 10's compliment and 123.123 is

(a) 876.876 , 877.876

(b) 876.876 , 876.877

(c) 987.987 , 988.987

(d) 987.987 , 987.988



Question

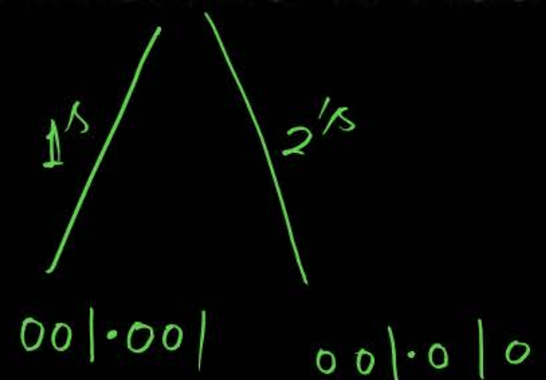
1's and 2's compliment of 110.110 will be :

(a) 001.001 , 010.001

(b) 001.001 , 011.001

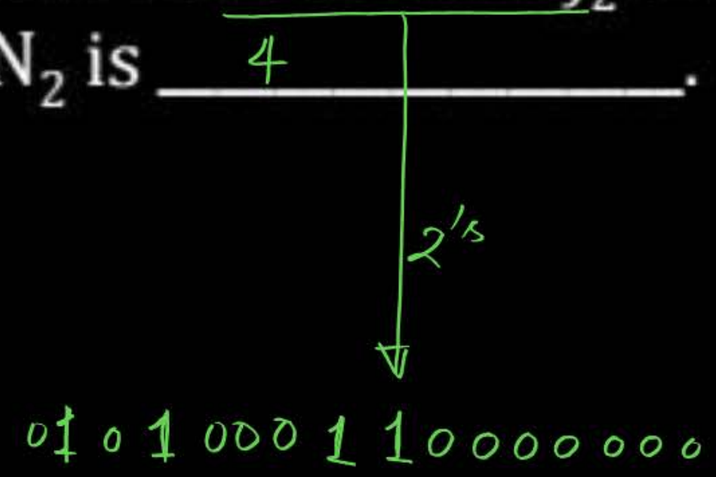
(c) 001.001 , 001.010

(d) 011.011 , 011.010



Question

A no. N_1 is given as $N_1 = (1010111010000000)_2$ the 2's compliment of N_1 is N_2. Total no. of 1's in N_2 is _____.



N_1 $\xrightarrow{1\text{'s}}$ N_2 $\xrightarrow{1\text{'s}}$ N_1

1011 $\xrightarrow{1\text{'s}}$ 0100 $\xrightarrow{1\text{'s}}$ 1011

$\xleftarrow{1\text{'s}}$

N_1 $\xrightarrow{2\text{'s}}$ N_3 $\xrightarrow{2\text{'s}}$

1011 $\xrightarrow{2\text{'s}}$ 0101 $\xrightarrow{2\text{'s}}$ $(1011)_2$

$\xleftarrow{2\text{'s}}$

$(785)_{10} \xrightarrow{9\text{'s}} (214)_b$

$\xleftarrow{9\text{'s}}$

$785 \xrightarrow{10\text{'s}} (215)_{10}$

$\xleftarrow{10\text{'s}}$

$(N_1)_n \xrightarrow{(n-1)\text{'s}} (N_2)_n$

$\xleftarrow{(n-1)\text{'s}}$

$(N_1)_n \xrightarrow{n\text{'s}} (N_3)_n$

$\xleftarrow{n\text{'s}}$

Question

2's compliment of no. N_1 is given as $(110.011)_2$ then no. N_1 is

(a) $(001.101)_2$

(b) $(001.100)_2$

(c) $(010.100)_2$

(d) $(001.011)_2$



$= 001.101$

Signed No. Representation

- Sign magnitude Method
- 1's compliment Method
- 2's compliment method

Positive no. is represented in same manner in all three representations.

Sign Magnitude Method

$n = 4$ bit

0 000	+0
0 001	+1
0 010	+2
0 011	+3
0 100	+4
0 101	+5
0 110	+6
0 111	+7

1 000	−0
1 001	−1
1 010	−2
1 011	−3
1 100	−4
1 101	−5
1 110	−6
1 111	−7

$n = 4$

-7 to 7

Total no. of distinct (different) numbers that can be represented using n-bit in sign magnitude method $= (2^n - 1)$

Range of the numbers represented from n-bits:

$$-(2^{n-1} - 1) \quad \text{to} \quad +(2^{n-1} - 1)$$

$(\underline{10|1|001})_2 = (-25)_{10}$

$(0|1|0|1|)_2 = +(27)_{10}$

1's Compliment Method

$n = 4$

0000 (+0)	$\xrightarrow{1's}$	$(1111)_2$ (−0)
0001 (+1)	$\xrightarrow{1's}$	1110 (−1)
0010 (+2)	$\xrightarrow{1's}$	1101 (−2)
0011 (+3)	$\xrightarrow{1's}$	1100 (−3)
0100 (+4)	$\xrightarrow{1's}$	1011 (−4)
0101 (+5)	$\xrightarrow{1's}$	1010 (−5)
0110 (+6)	$\xrightarrow{1's}$	1001 (−6)
0111 (+7)	$\xrightarrow{1's}$	1000 (−7)

-7 to $+7$

$(1010)_2 \xrightarrow{1's} (0101)_2$

$(-5)_{10}$ $+5$

$(10110)_2 \xrightarrow{1's} (01001)_2$

$(-9)_{10}$ $(+9)_{10}$

$A \xrightarrow{1's} -A$

Could be +Ve or −Ve

Total no. of distinct numbers that can be represented using n-bits in 1's complement representation $= (2^n - 1)$

Range of the numbers represented from n-bits :

$$-(2^{n-1} - 1) \quad \text{to} \quad +(2^{n-1} - 1)$$

2's Compliment Method

$(1000)_2 \xrightarrow{2's} (1000)_2$

$n = 4$ bits

0 000	(+0)	$\xrightarrow{2's}$	$(0000)_2$	(−0)
0 001	(+1)	$\xrightarrow{2's}$	$(1111)_2$	(−1)
0 010	(+2)	$\xrightarrow{2's}$	$(1110)_2$	(−2)
0 011	(+3)	$\xrightarrow{2's}$	$(1101)_2$	(−3)
0 100	(+4)	$\xrightarrow{2's}$	$(1100)_2$	(−4)
0 101	(+5)	$\xrightarrow{2's}$	$(1011)_2$	(−5)
0 110	(+6)	$\xrightarrow{2's}$	$(1010)_2$	(−6)
0 111	(+7)	$\xrightarrow{2's}$	$(1001)_2$	(−7)

$(1010)_2 \xrightarrow{2's} (0110)$, $(-6)_{10}$ ⟷ $+6$

$(11101)_2 \xrightarrow{2's} (00011)_2$, $(-3)_{10}$ ⟷ $(+3)$

$A \xrightarrow{2's} -A$

A → Could be +ve & −ve

$(\overset{-8\,4\,2\,1}{1010})_2 \longrightarrow (-8+0+2+0) = (-6)_{10}$

$(1110)_2 \longrightarrow -8+4+2 = (-2)_{10}$

−8 to 7

$$(1000)_2 = -8 + 0 + 0 + 0 = (-8)_{10}$$

- Total no. of distinct numbers that can be represented using n-bit in 2's complement representation $= 2^n$

Range of the numbers using n-bit

$$-(2^{n-1}) \quad \text{to} \quad +(2^{n-1} - 1)$$

Imp Discussion regrading 2's compliment representation

- $(1010)_2 = (-6)_{10}$

  $1\,1010 = -16+8+0+2+0 = (-6)_{10}$

  $11\,1010 = -32+16+8+0+2+0 = (-6)_{10}$

  $(0100)_2 = (+4)_{10}$

  $(0\,0100)_2 = (+4)_{10}$

  $(0\,0\,0100)_2 = (+4)_{10}$

$(1101)_2$

$=$

$\rightarrow (1111\ 1101)_2$

$(11101)_2$

$= (101)_2$

$= (-3)_{10}$

$$(11110101)_2 = (10101)_2$$

$$= -32 + 8 + 4 + 0 + 1$$

$$= (-19)_{10}$$

$(-13)_{10}$ using 8 bits in 2's compliment

5 bits → $(10011)_2$ → $(11110011)_2$

00001101

2's

2's

$(-27)_{10}$ → 8 bits → $(10010101)_2$

$(11100101)_2$

$(-7)_{10}$ → 8 bits →

$(1001)_2 = (1111\ 1001)_2$

Addition and Substraction in 2's compliment signed no. representation

Lets understand with example :

$n = 4$ bit

A+B

- $A = 0100 = (+4)_{10}$

  $B = 1011 = (-5)_{10}$

  $+ \quad 000$

  $1111 \longrightarrow$ Result $= 1111 = (-1)_{10}$

- $A = 0111 = +7$

  $B = 1110 = (-2)$

  $1\ 0101 \longrightarrow$ Result $= (0101)_2 = (+5)_{10}$

  (1 discard)

$$A - B = A + (-B)$$

$A = 0101 = (+5)$

$B = 0010 = (+2)$

$0111 \longrightarrow$ Result $= (0111)_2 = (+7)_{10}$

$A = 0011 = +3$

$B = 0010 = +2$

$0101 =$ Result $= (0101)_2 = (+5)_{10}$

- $A = 1011 = (-5)_{10}$

  $B = 1110 = (-2)_{10}$

  ① 1001 ⟶ Result $= (1001)_2 = (-7)_{10}$

  ← discard

- $A = 1101 = (-3)_{10}$

  $B = 1011 = (-5)_{10}$

  1

  ① 1000 ⟹ Result $= (1000)_2 = (-8)_{10}$

  ← discard

- $A = 0100 = +4$

  $B = 0101 = +5$

  1↑

  (0) 1001 ⟶ Result $= (1001)_2 = (-7)_{10}$

- $A = 1011 = (-5)_{10}$

  $B = 1100 = (-4)_{10}$

  0

  ① 0111 ⟶ Result $= (0111)_2 = (+7)_{10}$

  ← discard

- $A = 1010 = (-6)_{10}$

  $B = 1001 = (-7)_{10}$

  ① 0011 ⟶ Result $= (0011)_2 = (+3)_{10}$

  ← discard

Condition of Overflow

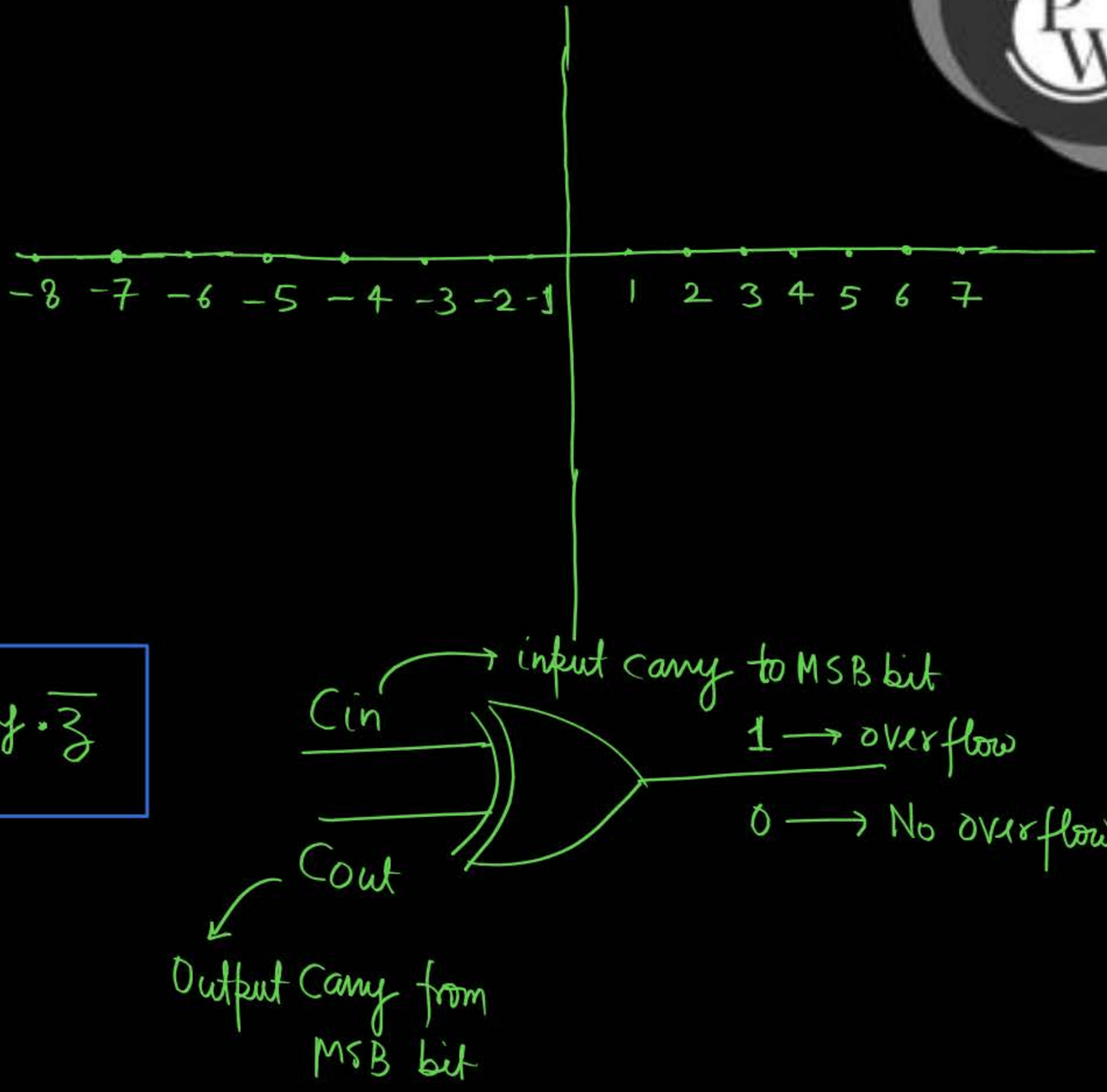
- Overflow may happen if and only if both the no. are of same sign.

$x \longrightarrow$ sign bit (MSB bit) of A

$y \longrightarrow$ sign bit (MSB bit) of B

$z \longrightarrow$ sign bit (MSB bit) of Result

$$\text{Overflow}[x,y,z] = \bar{x}\cdot\bar{y}\cdot z + x\cdot y\cdot\bar{z}$$



- $A = 0101$
  $B = 0111$
  $\rightarrow$ 2's compliment representation.

$A - B = ?$

$(B - A)$

$$A - B = A + (-B)$$

$$A = 0101$$
$$-B = 1001$$
$$+$$
$$(1110) \rightarrow \text{Result} = (1110)_2 = (-2)_{10}$$

$B = 0111 = (+7) \xrightarrow{2's} -B = (1001)_2 = (-7)_{10}$

$B - A \Rightarrow B + (-A)$

$A = 0101 \xrightarrow{2's} -A = (1011)_2 = (-5)_{10}$

$$B = 0111$$
$$-A = 1011$$
$$\leftarrow (1)\,0010 \rightarrow \text{Result} = (0010)_2 = (+2)_{10}$$

discard

Subtraction of unsigned No.s using 2's complement Method

$A = (1100)_2 = (12)_{10}$

$B = (0011)_2 = (3)_{10}$

$A - B \Rightarrow$ 1100

2's Comp of B + 1101

discarded ← (1) 1001 → final result $(1001)_2 = (9)_{10}$

Result is +Ve

$B - A \longrightarrow$ 0011

2's comp of A: 0100

0111 → Result $= -($2's comp of 0111$)$

No Carry → Result is −ve

$= -(1001)_2$

$= -(9)_{10}$

$A = 1011 = (11)_{10}$

$B = 0110 = (6)_{10}$

$A - B \longrightarrow A = 1011$

2's comp of B: 1010

discarded ← (1) 0101 → Result $= (0101)_2 = +(5)$

result is +Ve

$B - A =$ 0110

2's of A: 0101

1011 → Result $= -($2's comp 1011$)$

no Carry → Result is −Ve

$= -(0101)$

$= -(5)_{10}$

- n = 4 bit 2's compliment representation.

→ (A + B) ⟶ to avoid overflow ⟶ one extra bit in result will be required.

↓ −8 to 7

Result −16 to +14

if there are 5 bits for result ⟶ (−16 to 15)

↓ for n bits numbers if there is one extra bit i.e (n+1) bit in result then overflow can be avoided.

Sign magnitude Method :

$n = 4$ bit

$-(7)$ to 7

$(A + B)$

↓

Result → $(-14$ to $14)$

if we have 5 bits for result then we can represent $(-15$ to $+15)$ and can avoid overflow.

- So one extra bit is required for avoiding overflow in result.

1's compliment method :

$n = 4$ bit

-7 to 7

$(A + B)$

↓

Result -14 to 14

if we have one extra bit then we can represent -15 to 15 and can avoid the overflow.

- So one extra bit in result is required to avoid overflow.

Question

Two numbers A and B are represented using 2's compliment representation :

$A = (1011)_2$, $B = (1110)_2$

Then the results of (B – A) will be

H.W.

(a) $(0011)_2$

(b) $(10011)_2$

(c) $(1000)_2$

(d) $(1101)_2$



Topic : 2 Min Summary

Number System

Thank you

GW

Soldiers !

